

Periodic Trends

Answer key with worked solutions

A. Understanding

1. Effective nuclear charge.

The net positive charge felt by an outer electron after shielding by the inner electrons.
(a) Na: $11 - 10 = 1$ (b) S: $16 - 10 = 6$

2. Radius across a period.

Across a period protons are added, but the electrons go into the same outer shell and the number of inner (shielding) electrons stays the same. Z_{eff} increases, so the outer shell is pulled closer to the nucleus.

B. Ranking

3. Atomic radius.

In period 3, radius decreases from left to right ($\text{S} < \text{Mg} < \text{Na}$); Ca is below Mg with an extra shell. **$\text{S} < \text{Mg} < \text{Na} < \text{Ca}$**

4. First ionization energy.

Down group 1 ionization energy decreases ($\text{K} < \text{Na} < \text{Li}$); Ne, a noble gas at the right of period 2, is highest. **$\text{K} < \text{Na} < \text{Li} < \text{Ne}$**

5. Electronegativity.

$\text{Na} (0.93) < \text{I} (2.66) < \text{Br} (2.96) < \text{Cl} (3.16)$

C. Ionization energy

6. Group 1.

Ionization energy **decreases** down the group, from 520.2 kJ/mol (Li) to 375.7 kJ/mol (Cs). Each element has one more shell, so the outer s electron is farther from the nucleus and more shielded by inner electrons. It is held less strongly and is easier to remove.

7. Period 2 exceptions.

B < Be: boron's outer electron is in a 2p orbital, which is higher in energy than beryllium's 2s orbital, so it is easier to remove. **O < N:** nitrogen's three 2p electrons are unpaired; oxygen's fourth 2p electron must pair with another, and the repulsion between the paired electrons makes one of them easier to remove.

D. Ionic radius

8. Ions and atoms.

(a) Cl^- is larger: the extra electron increases repulsion between electrons, and the same 17 protons now hold 18 electrons. (b) **Mg** is larger: Mg^{2+} has lost its whole third shell ($3s^2$), and its remaining 10 electrons are held by 12 protons.

E. Challenge

9. Isoelectronic series.

$\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$. All have 18 electrons, but the number of protons increases (16, 17, 19, 20). More protons pull the same number of electrons in more tightly, so the ion is smaller.